

ORIGINAL ARTICLE

Detecting deception: employing deep neural networks for fraudulent review detection on Amazon

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Abstract

In the era of e-commerce dominance, an increase in fake reviews on online shopping platforms compromises the integrity of consumer feedback systems. This study focuses on Amazon, a leading e-commerce platform in the USA, where fake reviews have become a significant concern. Given the limited availability of authentic datasets for analysis, we propose a novel methodology to differentiate between genuine and fraudulent reviews across verified and non-verified purchases. Our approach utilizes the bootstrap distribution of cosine similarity values, providing a robust statistical foundation for review classification. We present a comprehensive framework integrating convolutional neural networks with word embedding and emotion-mining techniques through natural language processing, using a novel loss function. This multifaceted approach enhances detection accuracy and offers insights into the linguistic and emotional markers of fake reviews. Our method demonstrates exceptional performance, achieving an accuracy rate of over 96% in distinguishing fake reviews from user reviews. This study contributes to the growing research on online review authenticity and offers practical implications for e-commerce platforms, regulatory bodies, and consumers. This research aims to foster trust in online marketplaces and protect consumers from misleading information by providing a powerful tool for fake review detection.

Keywords Convolutional neural network · Natural language processing · Bootstrap · Cosine similarity · Amazon fake reviews

1 Introduction

In recent years, the popularity of online shopping has surged, leading to heightened competition among sellers driven by increasing consumer demand. In an effort to gain a competitive edge, some sellers have resorted to offering incentives such as gift cards, discounts, or complimentary products in exchange for positive reviews. These practices often result in biased and unreliable feedback, undermining the integrity of online reviews. Furthermore, certain sellers employ services or individuals to generate fake reviews in large quantities, thereby artificially inflating their product ratings. This manipulation frequently involves the creation of multiple fraudulent accounts to post positive reviews for their own products or negative reviews for those of competitors. Such deceptive practices ultimately mislead consumers by presenting overly favorable assessments, which can lead to dissatisfaction with the actual product quality and significantly damage consumer trust in both the seller and the platform.

The trustworthiness of online reviews plays a crucial role in consumer decision-making. According to Filieri, Trip Advisor users heavily rely on the content of travel reviews, with a particular focus on the degree of detail, the type of information provided, and the reviewer's writing style [1]. Lengthy reviews containing more comprehensive information are especially valued. Moreover, consumers who anticipate spending a substantial amount of

Both authors have contributed equally to this work.



money are likely to investigate reviews more thoroughly before making a purchase. On the other hand, Otero argues that enforcing legal action against false reviews is essential since fake reviews contribute to an unethical pool of advertising, which is detrimental to both competing sellers and consumers [2].

Mukherjee et al. discovered that fake reviews tend to be shorter than authentic reviews, based on data from Amazon Mechanical Turk (AMT) [3]. This finding aligns with the general assumption that fake reviews are typically more concise than genuine reviews. However, Martens and Maalej disagreed with this notion by demonstrating that in app store purchases, fake reviews are often lengthier than true reviews [4]. They also noted that fake reviewers tend to post more frequently than genuine reviewers. He et al. conducted a study identifying sellers who purchase fake reviews and examined the patterns in these reviews, along with their impact on overall rating changes [5]. Their findings revealed that fake reviews often include more photos, a short review title, and longer text. Additionally, the recent trend of acquiring fake reviews through Facebook groups and generating fake reviews using machine learning techniques has contributed to the creation of lengthier fake reviews.

Amazon is one of the largest online shopping platforms in the USA. According to Amazon Statistics, in 2023, US-based sellers sold more than 4.5 billion items, averaging approximately 8,600 items per minute [6]. These sellers also reported average annual sales exceeding \$250,000. According to Duarte et al. and Kaushik et al., customer reviews play a critical role in the online shopping experience, as the intangibility of products prior to purchase requires buyers to rely heavily on the experiences of others [7, 8]. Further, through a report by the Pew Research Center, Smith and Anderson noted that more than 80% of US customers frequently consult product reviews before making a purchase [9].

In recent years, Amazon has identified that fake reviews significantly impact product sales and consumer trust. According to Amazon, a troubling trend has emerged involving the “fake review broker” industry, where individuals or entities offer fake reviews in exchange for free products or monetary compensation [10]. This development has raised serious concerns about the authenticity of online reviews and their influence on consumer purchasing decisions.

Amazon’s review system differentiates between verified purchase (VP) and non-verified purchase (NVP) reviews to help users assess the reliability of feedback. To receive a VP label, reviewers must have purchased the product on Amazon and have spent at least \$50 on the platform within 12 months¹. This ensures the reviewer has a legitimate transaction history. VP reviews are considered more credible because they come from users with verified purchases. These reviews directly impact the product’s average rating. NVP reviews are written by users who have not purchased the product on Amazon or have not met the \$50 spending threshold. These reviews are initially not included in the product’s average rating unless additional details, such as text, images, or videos, accompany them. Although NVP reviews are not factored into the average rating without extra context, they can still be helpful if they provide genuine feedback. They are not necessarily fake, but Amazon requires additional information to ensure their inclusion in the overall rating. Both VP and NVP reviews can be authentic or misleading. VP reviews are not immune to bias or manipulation, while NVP reviews, even without purchase verification, can offer valuable insights if they are actual comments.

The structure of this paper is organized to provide a comprehensive exploration of fake review detection on online shopping platforms. We begin in Sect. 2 with a thorough literature review, examining the current state of research in this field and highlighting key approaches and methodologies employed by other scholars. Section 3 reviews the methodological approach that we apply in this study while stating the foundation of our study, detailing the data sources utilized and their specific characteristics. This section also outlines our particular focus on the process of labeling reviews as fake or genuine within our dataset. Further, this section expresses a modified loss function that we propose in order to boost the classification performances. The core of our research is presented in Sect. 4, where we conduct an in-depth analysis of the review data and assess the efficacy of a neural network model in detecting fake reviews. Finally, Sect. 5 synthesizes our findings, offering concluding remarks

¹ [UnderstandingCustomerReviewsandRatings:Amazon.com](https://www.amazon.com/UnderstandingCustomerReviewsandRatings).

on the implications of our research and its contribution to the broader understanding of fake review detection in e-commerce environments.

2 Related works

In online shopping platforms, the use of fake reviews by sellers to boost sales has become a pressing issue, prompting significant research into detection methods. Shojaee et al. tackled this problem by using manual annotation methods with metadata, where human annotators identified reviews with “Verified Purchaser” tags as genuine, while those without were flagged as potentially fake [11]. This approach highlights the importance of verified purchase indicators in assessing review authenticity. The rise of Electronic Word of Mouth (eWOM) has made online reviews crucial in consumer decision-making, but the increase in fake reviews threatens the credibility of these platforms. As a result, developing effective fake review detection techniques is essential to maintaining consumer trust and the integrity of online marketplaces.

Kokodis et al. and Anderson and Simester provide compelling insights into the nature and impact of fake reviews on online platforms, particularly Amazon [12, 13]. Kokodis et al. observed a significant decrease in product ratings following Amazon’s introduction of optional purchase verification, suggesting that non-verified reviews are more likely to be fake. This finding highlights the effectiveness of verification systems in curbing fraudulent review practices. Complementing this, Anderson and Simester’s research revealed that reviews without confirmed transactions often exhibit linguistic characteristics associated with deception. These studies collectively underscore the prevalence of fake reviews among non-verified purchases and demonstrate how verification mechanisms can help distinguish between genuine and potentially fraudulent feedback. Their findings emphasize the importance of implementing robust verification systems to maintain the integrity of online review platforms and protect consumers from misleading information.

Salminen et al. addressed the challenge of limited data availability for Amazon fake reviews by developing an innovative approach to fake review detection [14]. Their method involved generating fake reviews using two sophisticated language models: ULMFiT and GPT-2. By leveraging these models, they created a comprehensive dataset that combined machine-generated fake reviews with authentic human-written reviews. This synthetic dataset provided a foundation for training a machine learning model specifically designed to identify computer-generated reviews. The researchers’ approach not only overcame the scarcity of labeled fake review data but also enabled the development of more robust detection systems. Their work demonstrates the potential of using advanced language models to simulate fake reviews, thereby creating valuable training data for machine learning algorithms tasked with distinguishing between genuine and artificially generated content.

Sun et al. addressed the growing problem of fake reviews on online shopping platforms by proposing an innovative hybrid model for detection [15]. Their approach combines a convolutional neural network (CNN) with two support vector machine (SVM) models using a bagging ensemble technique. This method was developed and tested using the Amazon dataset from 2006, providing a robust foundation for analysis. The CNN component was designed to capture product-related review features, leveraging its ability to extract local patterns from text data. By integrating the CNN with two SVM models, which have proven effective in text classification tasks, the researchers aimed to enhance the overall accuracy of fake review detection. This hybrid approach demonstrates the potential of combining deep learning and traditional machine learning techniques to create more effective solutions for maintaining the integrity of online review systems.

Mukherjee et al. and Hajek et al. conducted significant research on fake review detection using diverse datasets and machine learning approaches [3, 16]. They utilized data from Yelp and Amazon Mechanical Turk (AMT), highlighting the challenges in distinguishing fake reviews from genuine ones when training models with AMT data. Hajek et al. employed advanced neural network architectures, specifically convolutional neural networks (CNNs) and deep feed-forward neural networks (DFFNNs), which is a multi-layer perceptron with two hidden layers [16]. These models incorporated n-gram, skip-gram, and emotional models to enhance their performance.

This hybrid approach yielded impressive results, achieving approximately 82% accuracy for Amazon data and around 90% accuracy for hotel and restaurant data. The integration of these diverse neural network architectures allows for a more comprehensive analysis of review content, capturing both local textual patterns (through CNN) and complex feature interactions (through DFFNN). This research underscores the potential of deep learning techniques in fake review detection while also emphasizing the importance of diverse and representative training data to improve model performance across different domains.

Barbado et al. and Zhang et al. presented innovative approaches to fake review detection, leveraging different aspects of review data and advanced machine learning techniques [17, 18]. Barbado et al. focused on analyzing Yelp data across various cities, utilizing Yelp's proprietary algorithm to label reviews as fake or genuine. Their key finding revealed significant differences in the mean values of user-centric and review-centric features between fake and non-fake reviews. This discovery led them to incorporate these distinguishing features into machine learning models, resulting in improved accuracy in fake review detection. Further, Zhang et al. proposed a more complex, multifaceted approach, employing the Longformer model for text embedding, which is particularly effective for processing long sequences of text [18]. They combined this with a hybrid neural network architecture, integrating convolutional neural networks (CNN) and bidirectional long short-term memory (BiLSTM) networks. Additionally, their model incorporated a behavior-sensitive feature extractor to analyze user behavior patterns and a context-aware mechanism to consider broader contextual factors. This comprehensive approach aimed to capture a more nuanced view of reviews, considering not just the text content but also user behavior and contextual elements, potentially offering a more robust method for identifying fake reviews.

Mohawesh et al. introduced a sophisticated semantic and linguistically aware model for the detection of fake reviews, achieving enhanced accuracy through the utilization of an advanced transformer-based architecture [19]. Their approach synergistically combined RoBERTa with an LSTM layer, enabling the model to discern complex patterns fundamental in fraudulent reviews and it enhanced the ability to distinguish fake reviews. Based on experimental evaluations, their model attained an accuracy of 96.03% on the OpSpam dataset and 93.15% on the Deception dataset.

Further, Zhang et al. proposed a composite architecture called DynDistilBERT-BiLSTM-CNN model dynamically allocates processing resources based on input complexity, combining sequential and local feature extraction to improve accuracy [20, 21]. Tested on the Amazon dataset, it outperformed models like ALBERT and RoBERTa, achieving up to 92.6% accuracy and demonstrating superior handling of detailed textual data through multifeature fusion.

Furthermore, Sherry et al. investigated the market for fake product reviews on Amazon.com, focusing on the purchase of reviews through private groups on Facebook and other platforms [22]. The researchers collected data on these markets and tracked the affected products' ratings, reviews, sales rank, advertising, and pricing on Amazon. Their findings reveal that a diverse range of products, including those with numerous reviews and high average ratings, engage in purchasing fake reviews. The practice of buying fake reviews on Facebook was found to result in a significant but short-lived increase in both average ratings and review counts. By analyzing a temporary policy change by Amazon, the study demonstrated that rating manipulation had a substantial causal effect on sales. Furthermore, the research explored whether this manipulation benefits consumers or was primarily used by high-quality products as a form of advertising or by new products to overcome initial visibility challenges. The results indicated that after companies cease buying fake reviews, their average ratings decline, and the proportion of one-star reviews increases notably, especially for newer products. This suggested that rating manipulation was predominantly employed by lower-quality products rather than serving as a legitimate marketing tool for high-quality items.

3 Data and methodology

Acquiring datasets with clear labels for fake and non-fake reviews on Amazon is difficult. Due to the nature of online reviews and the various methods used to identify fraudulent content, such datasets are often rare and not readily available. This makes it challenging for researchers and practitioners to access comprehensive and accurately labeled data to study and develop fake review detection methods. Most of the studies were based on Yelp data, which has a label with a fake status, AMT, or generated fake data using AI techniques [3, 14, 15, 17, 18]. As such, Salminen et al. used the publicly available Amazon Review Data (2018) dataset by Ni et al. to create fake reviews using ULMFiT and GPT-2 and used the Amazon review dataset as non-fake data to train the model [14, 23]. One drawback of this method is that their study assumed that the Amazon Review Dataset contains no fake reviews.

Our study also uses the Amazon 2018 dataset by Ni et al., which is an updated version of the Amazon review dataset released in 2014 and includes data from May 1996 to October 2018 [23]. We use the k-core dataset, which is a subset of the original extensive dataset and contains the overall rating, verified status, review time, reviewer ID, product ID, reviewer name, review, and summary. Although most of the researchers focus on specific product areas, we consider five datasets that show similar interest, namely Cell Phones and Accessories, Digital Music, Electronics, Industrial and Scientific, and Software.

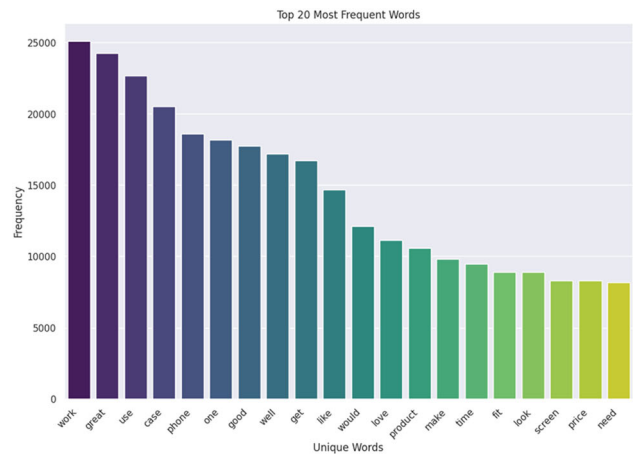
We construct our training dataset, which includes about two hundred thousand reviews across those five datasets. We select roughly balanced subsets from each original dataset while balancing the “verified status” feature. Since our focus is on a few similar product areas rather than a specific product area, this fusion from various datasets satisfies our interest.

Amazon has also identified some sellers who buy fake reviews through a group of reviewers with verified purchase signs. But those data are not publicly available. Consider a scenario in which a consumer has been an Amazon account holder for a long time and made purchases exceeding \$50 this year, which means the customer is a verified user of Amazon. However, if the customer has not purchased a product from Amazon and shopped from a different retailer, then the user tends to write a review for that product on Amazon; it will not show the sign of a verified purchase. But this comment is not fake. This highlights the challenge of treating NVP as fake reviews. Further, according to Amazon’s definition of VP and NVP, the purchases without VP are not all fake. Nevertheless, they do not count the ratings of these reviews for average rating calculation in some scenarios. All in all, these comments may be helpful for the consumers although the reviewer has not purchased the particular item through Amazon.

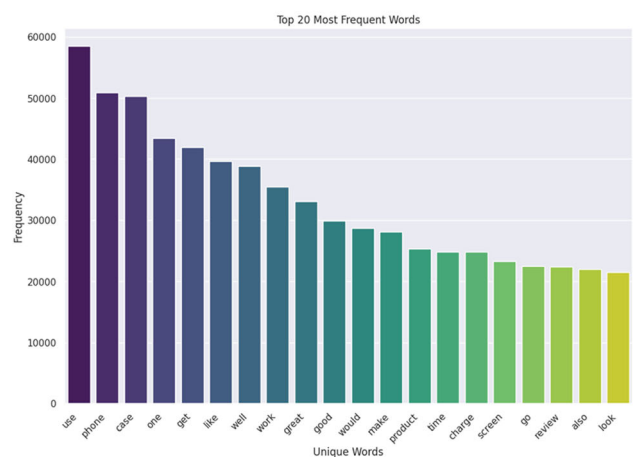
However, the original dataset contains the labels “verified purchase” and “non-verified purchase”, which is worth seeing the similarities in the comments between those two groups. The bar chart of the most frequent 20 words between the two groups is shown in Fig. 1. According to the figure, we can see that most of the words in those two groups are very similar and words like work, great, use, case, phone, one, get have very high frequencies in NVP group which suggests that in those two groups which supports the argument that the reviews with NVP label may contains some fake reviews which may be written by a fake reviewers or a machine. Also, there are some unique words such as one, love, price, and need in VP, and charge, go, and review in NVP, which also suggest that the NVP may contain some reviews that may not fake. It is hard to say that if a review has some single word which is more common in both VP and NVP groups, then that review in the NVP is fake. It is essential to read the entire review to identify the similarity between the reviews. As such, a reasonable method is required to modify the dataset with the fake review status label rather than treating all VP as non-fake and all NVP as fake.

Kauffmann et al. used the “Amazon Product Data by Julian McAuley 2014 ” dataset for sentimental analysis, and they considered the similarity between comments to label the reviews as fake reviews if the cosine similarity between reviews is greater than 0.85 [24]. In our study, we use a similar technique to label the reviews with fake and non-fake labels by addressing a few issues in this method, which is discussed in the methodology section.

Fig. 1 Relative frequency distribution of the most frequent words in the reviews for VP and NVP



(a) Reviews with VP label



(b) Reviews with NVP label

3.1 Methodology

Amazon is one of the most popular online retailers targeted by fake reviewers. Most of the studies are based on Yelp data because of the availability of fake labels. As such, in our study, we focused on Amazon's fake reviews. However, most of the available data does not contain the status of whether the review is fake or not, but it contains the label of verified status. According to Kokodis et al. and Simester, reviews with verified purchase labels are more trustworthy than reviews without verified purchase signs, which are highly likely to be fake [12, 13]. As such, it is necessary to modify the dataset with the fake status label of each review due to the scarcity of Amazon fake review data before train the model.

3.1.1 Term frequency-inverse document frequency (TF-IDF) weighting

TF-IDF measures the importance of a word to a document in a collection, adjusted for the fact that some words appear more frequently in general [25, 26]. The TF-IDF contains two statistics *idf* and *tf*. Inverse document frequency (*idf*) which measures how much information the word provides by considering how common or rare that word in all documents.

$$idf_i = \log\left(\frac{N}{df_i}\right)$$

where N is the total number of corpus in each document and df is the number of documents where the term T appears. Term frequency (tf) calculate the relative frequency of term T within document D . The TF-IDF weighting is given by

$$TFIDF_{i,j} = tf_{i,j} \times idf_i$$

3.1.2 Cosine similarity

Cosine similarity measures the similarity between two vectors using the inner product, which provides the angle between the two vectors [27, 28]. Recently, this method has been combined with TF-IDF [25, 26].

The cosine similarity between the two vectors is 1 if the angle between the two vectors is 0, such that those two vectors are precisely the same. Cosine similarity calculation is performed by the following formula between A and B two vectors:

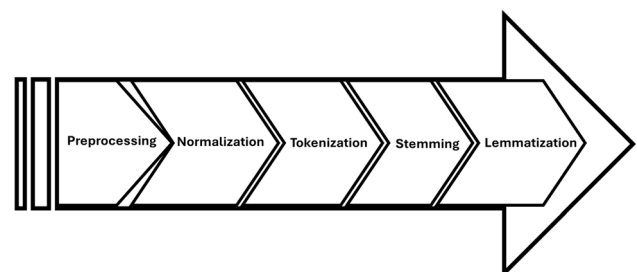
$$\text{Cos}\alpha = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i \times B_i}{\sqrt{\sum_{i=1}^n (A_i)^2} \times \sqrt{\sum_{i=1}^n (B_i)^2}}$$

3.1.3 Natural language processing (NLP)

Natural language processing (NLP) originated in the 1950 s, emerging from the combined realms of artificial intelligence and linguistic studies. Initially, NLP was distinct from text information retrieval (IR), which focused on efficiently indexing and searching large amounts of text using statistical methods. Today, NLP incorporates techniques and knowledge from a broad array of fields, which means contemporary NLP practitioners need to expand their expertise beyond traditional linguistic and computational domains. Issues like homographs—words that share the same spelling but have different meanings—and the challenge of translating metaphors led to notorious errors [29].

NLP is a text preprocessing pipeline that consists of four prominent steps, namely Preprocessing, Normalization, Tokenization, and Stemming and Lemmatization (see Fig. 2) [30–32]. Cleaning the text data by removing unnecessary elements such as symbols, HTML tags, URLs, hashtags, and emojis is a crucial step in simplifying our analysis. By stripping away these extraneous components, we ensure that the analysis focuses solely on the textual content that carries the intended meaning. This preprocessing step helps to enhance the performance of text analysis algorithms by reducing the complexity of the data, making it easier to identify patterns and draw conclusions. Therefore, this cleaning process is essential for ensuring that the analysis is accurate, relevant, and meaningful.

Fig. 2 Text preprocessing pipeline



Text normalization standardizes data by converting text to lowercase and correcting abbreviations, enhancing accuracy and efficiency in subsequent tasks [30, 33, 34]. Tokenization breaks text into individual units or tokens, typically words, enabling easier processing [35]. Using the right way to tokenize is essential for tasks like building language models, analyzing sentiment, or classifying text accurately [36, 37]. There are many different techniques for tokenization. Some are simple, like dividing text by spaces, while others are more advanced, such as tools like spaCy, NLTK, or HuggingFace's tokenizers. Stemming and lemmatization reduce words to their base forms, with stemming creating a simplified "stem" and lemmatization producing a valid "lemma" or dictionary form [30, 38]. The removal of stop words, which are common words with little semantic value, helps to reduce noise in the data [39]. These techniques collectively form the foundation for many NLP applications, allowing for more effective Comprehensive analysis of language by simplifying and standardizing text data while preserving essential meaning.

3.1.4 Sentiment analysis

Sentiment analysis is a computer-based method used in natural language processing (NLP) to detect, extract, and measure the emotional tone or attitude present in text. It simply identifies if the message expresses a positive, negative, or neutral tone [40]. There are different ways to perform this, including lexicon-based approaches, machine learning techniques, and combinations of both. Lexicon-based methods rely on predefined lists or dictionaries that assign sentiment values to individual words or phrases. On the other hand, machine learning-based analysis uses labeled data to train algorithms to recognize patterns of sentiment.

VADER, which stands for Valence Aware Dictionary and sEntiment Reasoner, is a tool that uses a list of words and rules to analyze feelings in text [41]. This tool is specifically built to handle social media updates, product reviews, and brief statements with ease. VADER assigns scores that show whether the text is positive, negative, neutral, or overall combined sentiment. It is recognized for its quickness and straightforwardness, but what really sets it apart is its ability to pick up on the detailed emotions conveyed through slang, emojis, and everyday phrases that people share online.

3.1.5 Word embedding

Word embeddings are a way of turning words into numerical vectors, which allows computers to grasp their meanings and how they relate to each other within a continuous space. Common methods like Word2Vec, GloVe, and FastText analyze large text collections to learn how words are connected based on context. These embeddings are essential for capturing similarities between words, solving analogies, and understanding language patterns in natural language processing tasks [42, 43].

GloVe embeddings, created by Stanford University, combine techniques that analyze entire matrices with methods that focus on local word environments [44]. They generate word vectors by studying how often words appear together across a large text collection. This approach helps these embeddings to effectively understand both the meaning and the grammatical relationships between words, making it easier for computer programs to understand and work with text more precisely.

3.1.6 Neural network (NN)

The phrase neural network (NN) brings to mind powerful images of machines that mimic the human brain, often evoking associations with science fiction and stories like Frankenstein. At the core of artificial neural networks are components known as artificial neurons [45]. A NN is a computational model inspired by the way biological neural networks in the human brain operate. It consists of interconnected layers of nodes, or "neurons" that work together to process and analyze complex data. Each neuron receives input from other neurons, processes this input using a mathematical function, and then passes the result to the next layer. NNs are particularly powerful

because they can learn from data, identify patterns, and make predictions or decisions without being explicitly programmed with specific rules [46–48].

Further, NNs are composed of three main types of layers: the input layer, hidden layers, and the output layer. The input layer receives raw data, which is then passed through one or more hidden layers where the network performs computations and learns the relationships within the data. The output layer produces the final result, such as a classification or prediction. The learning process involves adjusting the connections, or “weights,” between neurons based on the errors in the output, a method known as back-propagation [49]. This iterative process enables the network to improve its accuracy over time, making Neural Networks a foundational tool in fields like image recognition, natural language processing, and artificial intelligence.

3.1.7 Convolutional neural network (CNN)

Convolutional neural networks (CNNs) are similar to traditional NNs in their basic structure, as they are built from neurons that improve their performance through learning. Like in NNs, each neuron in a CNN processes inputs by applying operations such as scalar multiplication followed by a nonlinear transformation [50]. This foundational approach underlies many neural network architectures. In CNNs, however, the process involves transforming raw image data through multiple layers until a final output is generated, typically representing the likelihood of a particular class. The entire network functions as a single model that outputs a score based on learned weights. The final layer uses loss functions that relate to specific categories, and the strategies developed for optimizing traditional NNs, such as gradient descent and back-propagation, are also applicable to CNNs [50, 51].

Convolution operation, when f and g are two functions that can be discrete or continuous, can be written as equation 1,

$$(f * g)(X) = \begin{cases} \sum_{x=-\infty}^{\infty} f[x]g[X-x]\Delta x, & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} f(t)g(X-t) dt, & \text{if } X \text{ is continuous.} \end{cases} \quad (1)$$

where Δx is the sampling interval when X is discrete, while t is the time index of the output when X is continuous.

CNNs are specialized deep learning algorithms designed for processing structured grid data, particularly excelling in image-related tasks by leveraging spatial hierarchies [48]. Their architecture consists of convolutional layers for feature extraction, pooling layers for dimension reduction, activation functions for nonlinearity, and fully connected layers for final predictions [50, 51]. CNNs have become fundamental in deep learning, driving advancements in computer vision and finding applications in various fields. In NLP, CNNs are effective for tasks involving local pattern recognition, such as text classification and sentiment analysis, using word embeddings as input. However, their suitability for NLP tasks depends on the specific nature of the task, data, and model implementation, with architectural adjustments often necessary to manage computational complexity when dealing with numerous inputs [51–53].

3.1.8 Weighted binary cross-entropy

During the training of a neural network, the cost function plays a crucial role in refining the network’s weights, helping to develop a more accurate machine learning model. In the forward propagation phase, the network processes the training data, generating outputs that, for classification tasks, represent the predicted probabilities or confidence levels for various labels. These predicted probabilities are then compared to the actual target labels, and the loss function quantifies the error by assigning a penalty for any differences between the predicted and actual labels [54].

The binary cross-entropy loss function can be expressed as in equation 2,

$$E_b = -\frac{1}{N} \sum_{n=1}^N (y_n \times \log(g_\theta(x_n)) + (1 - y_n) \times \log(1 - g_\theta(x_n))) \quad (2)$$

where N represents the total number of training examples, y_n denotes the target label for the n^{th} training example, x_n corresponds to the input for the n^{th} training example, and g_θ refers to the model parameterized by the neural network weights θ .

In the subsequent back-propagation phase, the loss function's partial derivatives with respect to each trainable weight in the network are computed. These derivatives guide the adjustments to the weights. Through this iterative process of back-propagation, the neural network's weights are continuously fine-tuned, leading to a model that progressively minimizes the loss and improves its performance.

Imbalanced datasets challenge neural networks using standard cross-entropy loss functions, which treat all classes equally. To address this, a cost-sensitive cross-entropy loss function incorporating prior probabilities is proposed. This approach assigns higher weights to minority classes, increasing penalties for their misclassification. By making the loss function sensitive to class distribution, the neural network pays more attention to underrepresented data, improving its ability to predict rare events. This weighting strategy effectively balances the model's focus across all classes, enhancing overall performance on imbalanced datasets [55].

The weighted binary cross-entropy loss function applies distinct weights to the cost associated with incorrectly classifying positive examples and separately weights the cost for misclassifying negative examples. This approach allows for tailored penalization based on the type of error, reflecting real-world considerations where the consequences of false positives and false negatives may differ.

The weighted binary cross-entropy loss function can be modeled as in equation 3,

$$E_{wb} = -\frac{1}{N} \sum_{n=1}^N \left(c_{fn} \times y_n \times \log(g_\theta(x_n)) + c_{fp} \times (1 - y_n) \times \log(1 - g_\theta(x_n)) \right) \quad (3)$$

where N represents the total number of training examples, c_{fn} is the marginal cost of a false negative compared to a true positive, and c_{fp} is the marginal cost of a false positive compared to a true negative. The term y_n denotes the target label for the n^{th} training example, while x_n corresponds to the input for the n^{th} training example. The g_θ is parameterized by the neural network weights θ . By fine-tuning these weights, the model can achieve a better balance between sensitivity and specificity, ultimately leading to improved performance on imbalanced datasets.

3.2 Approach to detect fake status

Transforming the original dataset with VP and NVP labels into a binary classification problem of fake and non-fake labels presents significant challenges. Hence, in our study, we proposed a two-step modification to the dataset for discovering fake labels. Firstly, we modified the dataset by adding the “fake status” variable, which shows the fake and non-fake status of the review, to the analysis. To construct the “fake status,” we first detect fake users and label their reviews as “fake,” and, secondly, in the case of the remaining reviewers, we apply the cosine similarity method between VP and NVP to discover the fake and non-fake reviews. By comparing vectors in the review space, we can gain deeper insights into the similarities between different reviews [56]. This approach involves analyzing the vector representations of reviews, which capture various features such as content, sentiment, and length. By examining the similarities between these vectors, we can infer missing information by finding and utilizing patterns present in other reviews.

Upon doing the user analysis of the dataset containing counting NVPs and VPs per user, we discovered some users whose NVP reviews are way over the VP reviews, which is a clear indication that they are fake reviewers. Thus, all these NVP or VP fake writings by these reviewers must be assumed to be fake reviews.

Let m be the number of NVP reviews for a given reviewer and n be the total number of reviews by that reviewer. The condition for a particular reviewer being classified as fake if:

$$\frac{m}{n} > 0.5$$

If this condition is satisfied, all of the VP and NVP reviews are assumed to be fake from the corresponding user. In other words, reviewers that are strictly NVP with over 50% are considered fake. Therefore, the reviewers' VP and NVP were all assumed to be fake. As, for instance, the user has two reviews such that one is VP and the other is NVP, does not count as fake. Amazon recently identified the fake review broker industry, which posts fake reviews as verified purchases that are surging. Although we do not have data with that information, we assume that by using the condition for a particular reviewer being classified as fake, the NVP percentage of each user captures the VP reviews that are fake in the dataset.

As the second step, we employ a novel approach to identify fake reviews from the remaining reviews. However, we identified some issues with Kauffmann et al.'s research methods: i) Why do we use 0.85 as the threshold? Is there any specific reason to choose 0.85? ii) If two reviews are closer, which review will be labeled as fake? which are not clearly mentioned in their study [24]. The first issue pertains to determining an appropriate cosine similarity threshold value for identifying fake reviews in NVP. Salminen et al. utilized the GPT-2 model to generate fake reviews by fine-tuning it with Amazon data from 2014, thus creating a dataset containing both fake and genuine reviews [14].

Following this methodology, we generate 500 computer-generated reviews using the GPT-2 model, which, unlike other language models such as GPT-3, GPT-4, LLaMA, and Claude, GPT-2 is completely open-source and easily accessible to the public, which allows us to fully disclose and replicate our experiments [57]. GPT-2's proven capability to produce convincing synthetic reviews is well-documented; for example, Salminen et al. demonstrated that reviews generated by GPT-2 closely resemble genuine deceptive reviews in terms of tone, vocabulary, and overall structure [58]. Moreover, GPT-2 is notably computationally efficient, requiring markedly fewer resources than more advanced counterparts [59]. This increased accessibility not only encourages broader participation in related research but also aligns smoothly with the objectives of our study.

We create a pool of fake reviews by using non-fake VP (VP_{Nfk}) reviews from the Amazon 2018 dataset [60]. The next step involves calculating the cosine similarity between each GPT-2-generated fake review and its corresponding original Amazon VP review. For each GPT-2-generated fake review (GPT_i) and its corresponding non-fake VP review ($VP_{Nfk(i)}$), the cosine similarity $Cos(\alpha_i)$ between the two vectors is calculated as follows:

$$Cos(\alpha_i) = \frac{GPT_i \cdot VP_{Nfk(i)}}{\|GPT_i\| \|VP_{Nfk(i)}\|},$$

where $GPT_i = (G_1, G_2, \dots, G_{500})$ is the vector representation of the i^{th} GPT while $VP_{Nfk(i)} = (V_1, V_2, \dots, V_{500})$ is the corresponding vector representation of the i^{th} VP review. Further, $\|GPT_i\|$ and $\|VP_{Nfk(i)}\|$ are the Euclidean norms (magnitudes) of the corresponding vectors.

Therefore, $Cos(\alpha_i)$ provides a collection of five-hundred cosine similarity values that can be used to generate a bootstrap distribution of the cosine similarity values for the fake reviews. For each bootstrap sample b , we can calculate the sample mean via,

$$\bar{s}^{(b)} = \frac{1}{n} \sum_{i=1}^n s_i^{(b)},$$

where $s_i^{(b)}$ is the B number of bootstrap samples $\{cos(\alpha_1)^{(b)}, cos(\alpha_2)^{(b)}, \dots, cos(\alpha_{500})^{(b)}\}$ for $b = 1, 2, 3, \dots, B$, where each sample is drawn with replacement from the original set $\{cos(\alpha_1), cos(\alpha_2), \dots, cos(\alpha_{500})\}$.

To establish a threshold for identifying fake reviews, we calculated the 95% lower confidence bound of this cosine similarity distribution. The threshold value, denoted as k , serves as a critical parameter in distinguishing between fake and genuine reviews in the NVP dataset as follows,

$$Label_i = \begin{cases} NVP_{FakeReview}; & \text{if } cos(\theta_i) > k \\ NVP_{Non-FakeReview}; & \text{Otherwise} \end{cases}$$

where $cos(\theta_i)$ is the cosine similarity between VP_{NFK} reviews and NVP reviews. By setting this threshold, we aimed to achieve a balance between minimizing false positives and maximizing the accurate identification of fake reviews.

We believe that it is not reasonable to remove both reviews as mentioned in Kauffmann et al. study, because one of the reviews can be a true review, and the other one may be recreated by machine or human by coping original review. Therefore, instead of comparing the entire review pool with each other, we compare each NVP review with VP_{NFK} review. If the cosine similarity value is greater than the optimal threshold value(k), label the non-verified purchase review as a “fake review,” and the review with a verified purchase sign remains as “non-fake review”.

On the other hand, fake review detection using textual features has only become challenging as fake reviewers adapt methods to write reviews that are close to non-fake reviews. To address that issue, we refer the feature-centric analysis that has been done by Fontanarava et al. for Yelp data using supervised classification with review and reviewer-centric features [61]. It is hard to derive more features in the Amazon dataset because it has limited variables. Our model is trained using the modified data by including additional features such as rating, length of the review, and sentiment score of the review. The Amazon dataset has 29 categories, and we consider Cell Phones and Accessories, Digital Music, Electronics, Industrial and Scientific, and Software categories only. We select stratified random samples from each category to create a training dataset and used the remaining data of each category to test the accuracy of the model. Then, we use CNN to train the predictive model while splitting the dataset into 70:30 percentage ratio for training and testing, respectively.

Our CNN model consists of three convolutional layers that process data one after another to detect increasingly complex and abstract patterns within the text. After each layer, a pooling step is performed, which simplifies the data and emphasizes the most important features. This process helps the model better understand new or unfamiliar data. The first layer uses 128 filters to capture basic text features, such as simple keywords or patterns. The second layer uses 64 filters to recognize more intermediate features, like phrases or parts of speech. The third layer uses 32 filters to capture more complex and abstract patterns in the text. These layers work together to help the model understand different depths of information, starting from basic details and building up to more complex concepts. Employing several convolutional layers allows the model to develop a layered understanding of the text, making it easier to detect detailed language patterns that could indicate a review is fake or deceptive.

In our design, all three convolutional layers used a kernel size of 5. We specifically chose this size because it helps the model recognize patterns of five words at a time, capturing important meaning and sentence structure, which aids in distinguishing real reviews from fake ones. Also, we decreased the number of filters from 128 in the first layer to 32 in the last, which makes the model faster and less likely to memorize the training data. This setup allows the network to start by detecting broad features in the text and gradually focus on more specific, detailed patterns. We chose the rectified linear unit (ReLU) as the activation function for each convolutional layer because it is simple to use and highly effective. ReLU helps prevent the problem of vanishing gradients, allowing the

model to learn more quickly and stay stable during training. Also, ReLU adds the necessary nonlinearity to the network, which is important for understanding complex patterns and details in text data.

After each of the first two convolutional layers, a MaxPooling operation with a pool size of 2 was performed. MaxPooling serves to shrink the size of the feature maps, emphasizing only the most major features within each pooling window. This process helps to lower the computational requirements and memory consumption, while also making the model more resilient to small changes in the input data. After the third convolutional layer, a GlobalMaxPooling1D layer was applied. This layer reduces the entire feature map to a single vector by choosing the highest value from each feature map, resulting in a compact, fixed-length summary of the review text. Such a representation is especially helpful in emphasizing the most important overall features relevant for classifying text. To help the neural network generalize better and prevent it from overfitting the training data, we incorporated a technique called Dropout right after the first fully connected layer. We set the dropout rate to 0.5, which means that during each training cycle, half of the neurons are turned off at random. This approach makes the network less dependent on specific neurons or features, promoting more reliable performance and increased robustness across different data sets. After passing through the convolutional and pooling layers, the output was combined with extra information such as user ratings and sentiment scores to include important numerical details relevant to the context. Next, a fully connected layer with 64 units and a ReLU activation function was used to capture detailed patterns from the mixed textual and numerical data. The model applied a final sigmoid activation to distinguish between genuine and fake reviews, resulting in a straightforward yes-or-no classification. The model is compiled using an optimizer and a custom loss function, which is hybrid adaptive cross-entropy (HACE), along with accuracy and metrics.

3.2.1 Hybrid adaptive cross-entropy (HACE)

The hybrid adaptive cross-entropy (HACE) loss function is a refined version of the weighted binary cross-entropy loss function (4). It incorporates two key modifications to enhance its effectiveness. First, HACE employs dynamic class weights that adapt to the current state of the model, replacing the fixed weights c_{fn} and c_{fp} used in traditional approaches. Second, it integrates a focusing parameter γ , drawing inspiration from the focal loss concept. This parameter serves to assign greater importance to challenging classification instances, thereby improving the model's ability to learn from difficult examples.

The novel HACE loss function can be stated as,

$$E_{HACE} = -\frac{1}{N} \sum_{n=1}^N \left[w_n \left(y_n (1 - \alpha) + \frac{\alpha}{2} \right) (1 - g_{\theta}(x_n))^{\gamma} \log(g_{\theta}(x_n)) + (1 - w_n) \left((1 - y_n) (1 - \alpha) + \frac{\alpha}{2} \right) (g_{\theta}(x_n))^{\gamma} \log(1 - g_{\theta}(x_n)) \right] \quad (4)$$

where N represents the total number of training examples, w_n is the dynamic the marginal cost of a false negative compared to a true positive, α is a label smoothing parameter (small value) to prevent overconfidence, and γ is a focusing parameter. The proposing E_{HACE} loss changes with respect to the α and γ parameter tuning can be seen in the Appendix.

In the HACE loss function, the weighting factor is adjusted dynamically to reflect the additional cost of misclassifying a false negative compared to a true positive. Unlike the fixed class weights often used in standard binary cross-entropy, this factor is calculated on the fly, taking into account the changing distribution of training data and the model's current prediction performance at each training epoch. Specifically, at the end of each training epoch, w_n is recalculated as follows:

$$w_n = N_{negative}/N_{positive},$$

where $N_{negative}$ and $N_{positive}$ denote the number of negative and positive examples, respectively, correctly identified by the model during the current epoch. If the model tends to incorrectly label positive cases (leading to more false negatives), the value of w_n increases accordingly. This raises the penalty for false negatives in the next round of training. Such an energetic method helps the model to improve itself by paying more attention to difficult or less common examples during training, promoting better learning and accuracy.

Label smoothing (α) is a technique that subtly adjusts the target labels in the loss function to reduce the risk of a model becoming too confident in its predictions. Instead of using strict 0 s and 1 s, label smoothing slightly distributes some probability mass to other classes, promoting more cautious predictions and better generalization. After thorough cross-validation, we settled on a smoothing parameter alpha of 0.1 for this study (see Appendix). This value strikes a good balance, helping the model distinguish classes accurately while also avoiding over-confidence, thereby improving its ability to perform well on new data. The focusing parameter (γ), inspired by the focal loss approach, is customized to make the model more attentive to difficult classification cases, especially those that are misclassified with low confidence. Raising gamma causes the model to focus more on the difficult-to-predict examples, encouraging it to perform better on uncertain or borderline cases. In our study, we conducted extensive hyperparameter tuning through grid search, which led us to find that setting gamma to 2 achieved the best overall trade-off. This value effectively helps the model focus on challenging samples while ensuring stable and reliable performance.

Furthermore, having separate testing datasets are crucial for evaluating the model's performance accurately and generalize. It provides an objective evaluation, as the model hasn't seen the test data during training, thus reflecting its real-world capabilities. This practice helps prevent overfitting, where the model learns the training data too well, capturing noise that doesn't generalize. Additionally, using a separate validation set for hyperparameter tuning and model selection ensures that the test set remains unbiased, providing a fair benchmark for comparison. This approach also prevents data leakage, where unintended information from the training process influences the model, leading to overly optimistic performance metrics. Overall, testing on unseen data provides reliable performance metrics, ensuring the model's true abilities are understood and any improvements are genuinely reflected.

Moreover, the presence of outliers significantly impacts the model's performance. However, the decision to eliminate these anomalies requires strong justification. In our approach, we choose to retain all data points, including potential outliers, due to the absence of compelling reasons for their removal. This decision ensures that we maintain the integrity of the dataset and avoid potentially arbitrary exclusions that could affect the model's overall reliability and generalizability.

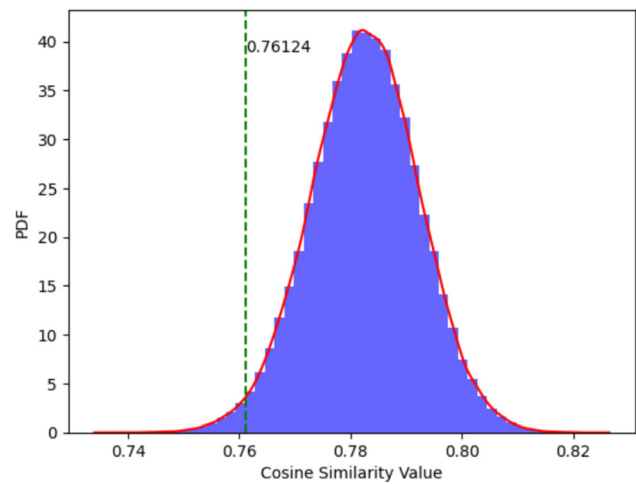
4 Results

4.1 Exploratory data analysis

4.1.1 Cosine similarity distribution of fake reviews

The bootstrap distribution of the mean cosine similarity was obtained by having 100000 bootstrap samples of sample size 500. The sampling distribution of the mean cosine similarity of fake reviews is symmetric with mean around 0.784 and standard deviation 0.009 (see Fig. 3). By using the bootstrap distribution, we obtained the 95% Bootstrap lower confidence bound for the mean cosine similarity which is 0.76124 which suggests that the 95% of the mean cosine similarity values between machine-generated reviews and VP reviews in the original dataset is greater than 0.76124. Therefore, the cosine similarity threshold value $k = 0.76$ is used to create the fake review label in the dataset. So, any review with NVP label has a cosine similarity ≥ 0.76 is considered as a fake review.

Fig. 3 Bootstrap distribution of Cosine similarity values



4.1.2 Ratings, user and review analysis of fake and non-fake reviews

When we look at the structure of the modified dataset, that means once we labeled them as fake and non-fake, it consists of 44.34% users only with fake reviews, 54.09% users only with non-fake reviews, and 1.57% users with both fake and non-fake reviews.

Firstly, a word cloud analysis is conducted to visualize the similarity between the words used in fake and non-fake reviews. The analysis revealed that certain words, such as “use,” “case,” “phone,” “product,” “one,” “really,” “great,” and “well” appeared frequently in both categories (see Fig. 4). This finding suggests that fake reviews often contain words similar to those found in non-fake reviews, complicating the task of distinguishing between the two based solely on word frequency. This outcome aligns with our initial expectation that differentiating between fake and non-fake reviews would be challenging due to the overlap in commonly used words.

Secondly, we analyzed the distribution of user ratings in both fake and non-fake reviews, as illustrated in Fig. 5. The analysis shows that five-star ratings are the most common in both fake and non-fake review categories, followed by four-star ratings. Notably, approximately 60% of the five-star ratings in the fake review category are indeed fake, while more than 60% of the five-star ratings in the non-fake category are genuine. In contrast, the frequency of fake reviews is higher for ratings from one to four stars compared to non-fake reviews. This pattern suggests that while fake reviews often aim to provide high ratings to influence product perception, non-fake reviews are more evenly distributed across different rating levels. Consequently, this observation suggests that the rating feature is a valuable indicator for distinguishing between fake and non-fake reviews. Therefore, it is reasonable to include rating as a key variable in our analysis, as it provides a discernible pattern that can aid in the identification of potentially fake reviews.

Some researchers have suggested that fake reviews tend to be longer than non-fake reviews. To investigate whether this holds true for our dataset, we analyzed the length distribution of the reviews. As shown in Fig. 6, both distributions are skewed, with fake reviews generally being longer than non-fake reviews. For better

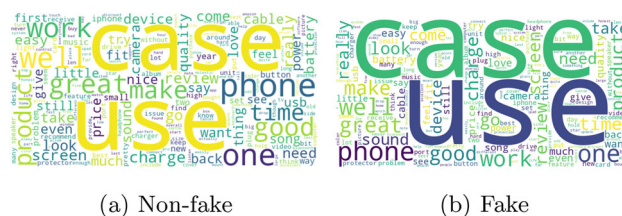
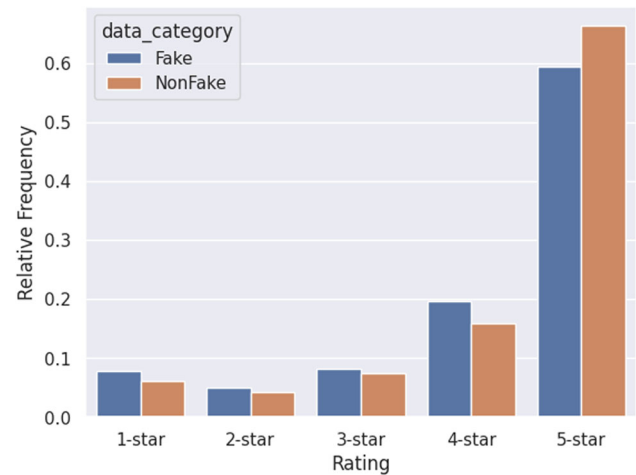
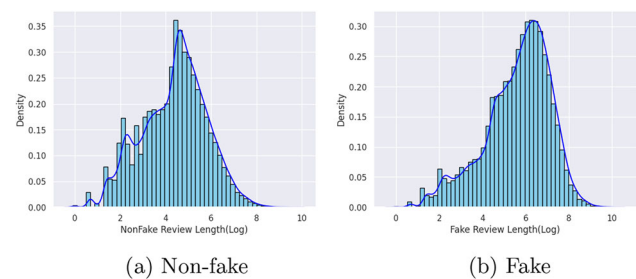
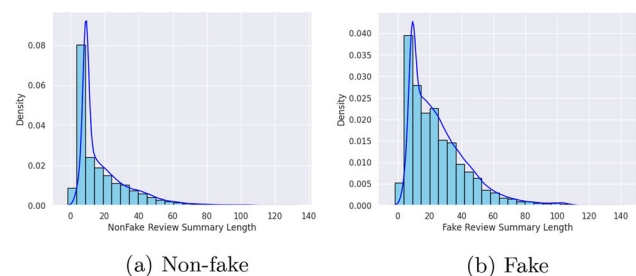


Fig. 4 Word clouds of fake and non-fake reviews: The major similarity in commonly used phrases shows that fake reviews often imitate genuine ones, making it hard to tell them apart just by looking at keywords

Fig. 5 Bar chart of ratings**Fig. 6** Length(Log) distribution of non-fake and fake reviews

visualization, we used a logarithmic scale for review length. The median word count for non-fake reviews is 95, while for fake reviews it is 367. This finding indicates that longer reviews are more likely to be fake. Additionally, there are a few anomalies in the dataset, with some reviews, both fake and non-fake being unusually long. We retained these reviews in our study because, despite their length, they provide valuable information similar to shorter reviews. Thus, review length emerges as a useful feature in identifying fake reviews in our analysis, supporting the hypothesis that review length can be a distinguishing factor between fake and non-fake reviews.

The length of the summary (or title) for each review was also considered, and the distribution of summary lengths is depicted in Fig. 7. Both distributions are skewed to the right. The median summary length is 13 words for non-fake reviews and 22 words for fake reviews. The distribution of non-fake summary lengths exhibits a sharp decline after the peak, suggesting that most non-fake reviews have relatively short summaries. This implies that non-fake reviews often provide concise summaries. In contrast, the distribution of fake summary lengths also shows an initial sharp peak, but the decline is less steep compared to non-fake reviews. This indicates a slightly more varied length of summaries in fake reviews. Overall, fake reviews tend to have longer summaries than non-fake reviews. This characteristic can be utilized as a feature in our analysis, as it provides another dimension for distinguishing between fake and non-fake reviews.

Fig. 7 Summary length distribution of non-fake and fake reviews

The sentiment score of each review was calculated to identify the emotional tone conveyed by the review. In our study, we employed VADER to perform sentiment analysis on reviews of Amazon products. Each review was evaluated on its own, with VADER assigning scores for positive, negative, and neutral sentiments, and an overall (compound) score that ranges from -1, indicating highly negative sentiment, to +1, reflecting strongly positive feelings. The compound scores were especially useful because they offered a quick summary of the sentiment expressed in each review. We then used these scores as features in our predictive model to help identify whether a review was authentic or fake. The distribution of sentiment scores for both fake and non-fake reviews is shown in Fig. 8. Both distributions are skewed to the left, with a few spikes indicating high frequencies in both categories. Most fake reviews have sentiment scores closer to 1, indicating predominantly positive feedback, especially when compared to non-fake reviews.

The median sentiment score is 0.637, which means more neutral, for non-fake reviews and 0.875, which means more positive, for fake reviews. The distribution of sentiment scores for fake reviews shows a strong concentration of highly positive sentiments, while the distribution for non-fake reviews is more varied, with significant densities across different points in the range. This pattern suggests that fake reviews are more likely to display extremely positive sentiment scores, whereas non-fake reviews exhibit a broader spectrum of sentiments, ranging from positive to neutral. This distinction in sentiment score distributions can be a valuable feature for distinguishing between fake and non-fake reviews.

Furthermore, some features may not demonstrate significant associations with our categorization, leading us to exclude them from the model. To evaluate the relevance of these features, we utilized a statistical feature selection method known as “SelectKBest” from the scikit-learn library for our univariate feature selection process in Python. This method employs the *f*_classif as its scoring function, which assesses the ANOVA F-value between features for classification tasks and ranks them by their importance [62]. The “SelectKBest” function in Python was utilized to identify features that are likely independent of the category and thus irrelevant for our classification. Upon reviewing the ranked list of features, we observed a substantial difference between features (as shown in Table 1). This clear distinction leads us to more focus on a few features with the highest scores, as these were deemed the most significant for our model. The scores represent how features relevant to the final target.

Based on the results, it is evident that several features are beneficial for distinguishing between fake and non-fake reviews. These include the length of the review, the sentiment score of the review, the comparison between the rating and the sentiment score of the review, and the sentiment score of the summary. However, since the rating and the comparison between the rating and the sentiment score provide significant insight into the review’s nature, it is not necessary to include the sentiment score of the title as an additional feature. With these carefully selected features, we can train a predictive model to classify reviews as fake or non-fake. The model will analyze each review’s text, leveraging the identified features to make accurate classifications. This approach ensures that the model is both efficient and effective in identifying fake reviews, as it focuses on the most relevant and informative aspects of the review data.

Fig. 8 Sentiment score distribution of non-fake and fake reviews

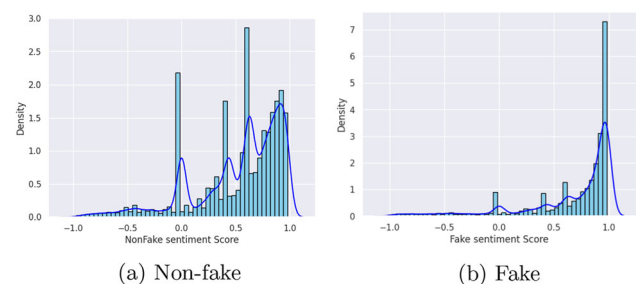


Table 1 Ranking important features in the dataset using the “SelectKBest” function

Feature	Score	<i>p</i> -value
Review Length	16556.34	< 0.000
Sentiment Score (Review)	4625.37	< 0.000
Rating vs. Sentiment Score (Summary)	1191.86	< 0.000
Sentiment Score (Summary)	709.24	< 0.000
Rating	549.78	< 0.000
Rating vs. Sentiment Score (Review)	100.91	< 0.000

4.2 Model performance

Our goal is to automate the classification of identifying fake and non-fake reviews from Amazon product reviews. To achieve this, we employ CNN, a commonly used type of deep neural network, to build a predictive model. The CNN model is trained using embedding features such as rating, review length, sentiment score of the review, and sentiment score of the summary. After training the CNN model, we test and five-fold cross validate the model using 30% of the test data as we setup.

In order to determine if the CNN method is the best option for our data, we evaluate its performance and execution time by comparing it with a couple of other transformer-based encoders, using a subsample of 15,000 data points (see Table 2). Although transformer-based models like DistilBERT and RoBERTa are designed to capture long-distance relationships within texts, our test on a subset of 15,000 samples shows that a convolutional neural network actually outperforms these models in all key evaluation metrics. As detailed in Table 2, the CNN achieves the highest F1-Score (0.936 compared to 0.915 and 0.905 for the transformers) and the best ROC-AUC score (0.954 versus 0.942 and 0.940). The Recall score of 0.942 shows that the model is highly effective at correctly identifying true positive cases. Also, the CNN processes data considerably faster, being 4.3 times quicker than DistilBERT and 2.9 times faster than RoBERTa (taking 112.41 s compared to 478.81 and 325.20 s). This emphasizes its efficiency in terms of computation.

In our study, we used GloVe word embeddings that were pre-trained on a large dataset consisting of 6 billion tokens from common crawl data, with each embedding having 100 dimensions. These embeddings were incorporated into our convolutional neural network model to improve its ability to understand the meaning behind the reviews. Each review was changed into a series of embedding vectors, allowing the CNN to detect important semantic patterns and language structures that differentiate fake reviews from genuine ones.

On top of that, we used the NLTK TweetTokenizer for tokenizing text, as it is specifically designed to handle informal and casual language often found in social media posts, online reviews, and quick messages. This tokenizer is particularly effective at managing common features seen in Amazon reviews, such as emojis, emoticons, slang words, punctuation marks, and abbreviations. These elements are essential for accurately analyzing sentiment and content. Each review was divided into a series of tokens that preserve key details, helping improve embedding quality and leading to more accurate model training.

Table 2 Performance comparison metrics. CNN outperforms on all relevant metrics after only a training epoch and does not change for the rest of 10 epochs

Architecture	Time (s)	Precision	Recall	F1-Score	ROC-AUC
CNN	112.41	0.931	0.942	0.936	0.954
DistilBERT	478.81	0.92	0.91	0.915	0.942
RoBERTa	325.2	0.91	0.9	0.905	0.94

We employ various loss functions to train our model and compare their performance to identify which one best represents the data. The comparison includes loss functions such as hybrid adaptive cross-entropy, weighted binary cross-entropy, Tversky loss, binary cross-entropy, and F1 loss, all of which are integrated within the CNN model. The results of this comparison are presented in Table 3.

Each loss function in Table 3 is designed to optimize the model's performance in different ways. Binary cross-entropy is a standard loss function for binary classification problems, focusing on the overall accuracy. F1 loss emphasizes the balance between Precision and Recall, making it particularly useful when both false positives and false negatives are critical. Tversky Loss is a generalization of the Dice coefficient, providing flexibility in handling imbalanced datasets by adjusting the penalty for false positives and false negatives.

The flexibility offered by weighted binary cross-entropy in adjusting the class weights ensures that the model is not overwhelmed by the imbalance in the data. For detailed parameter adjustment of the weights for the positive and negative classes through parameters α and β and their effects, please refer to Appendix.

However, among all the loss functions evaluated, hybrid adaptive cross-entropy outperforms the others (see Appendix for a graphical view of loss function comparison). This superior performance is attributed to its ability to handle imbalanced data more effectively and smoothly. Imbalanced data, where the number of fake reviews is significantly different from the number of non-fake reviews, pose a challenge for standard loss functions. The HACE addresses the imbalanced class problem by dynamically tuning its α and γ parameters (see Appendix). In other words, we can emphasize the minority class (fake reviews) and reduce the bias toward the majority class (non-fake reviews) without adding a fixed weight, thus enhancing the model's ability to correctly classify fake reviews.

Also, our ablation studies clearly demonstrated the critical role of engineered features, such as sentiment scores and user ratings. Combining these features with the text content consistently led to better classification results, far outperforming methods that relied solely on textual data (see Table 4). These findings emphasize the important impact of our innovative loss function and the carefully designed features in accurately identifying fake reviews.

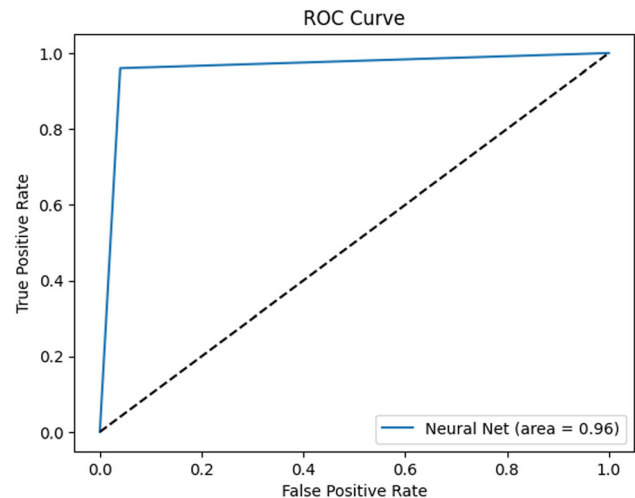
Therefore, the predictive CNN model demonstrates strong performance, achieving an overall accuracy with an ROC-AUC score of 0.961 suggests that it is highly effective at correctly identifying fake and non-fake reviews, achieving near-perfect classification accuracy (see Fig. 9). Additionally, the model exhibits high Precision at 0.96, Recall at 0.94, and an F1-Score of 0.95 in predicting fake reviews. A Precision score of 0.96 of the reviews the model identified as fake, is actually fake. This high precision indicates that the model has a low rate of false positives, meaning it rarely misclassifies genuine reviews as fake. A Recall score of 0.94 means that the model correctly identifies 94% of all fake reviews in the dataset. This high recall indicates that the model has a strong ability to detect fake reviews, with a relatively low rate of false negatives (failing to identify actual fake reviews). An F1-Score of 0.95 indicates a strong overall performance, balancing the trade-off between Precision and Recall. This means the model is both accurate in its positive predictions and comprehensive in identifying all relevant instances. Overall, these metrics indicate that the CNN model is highly effective in distinguishing between fake and non-fake reviews, providing reliable and accurate predictions. The high Precision ensures that most of the identified fake reviews are truly fake, while the high Recall ensures that most actual fake reviews are correctly detected. The combination of these metrics, reflected in a strong F1-Score, confirms the model's suitability for automating the classification of fake reviews in various product categories.

Table 3 Ablation study on loss functions

Loss function	ROC-AUC	Precision	Recall	F1 score
Binary cross-entropy	0.943020	0.926788	0.966846	0.946393
Weighted binary cross-entropy	0.945008	0.841678	0.880976	0.860874
F1 Loss	0.938674	0.838140	0.865122	0.851417
Tversky loss	0.940607	0.673407	0.816327	0.738011
HACE loss	0.961528	0.951496	0.970449	0.960879

Table 4 Ablation study on engineered features (F1 Score)

Feature set	Binary cross Entropy	F1 Loss	HACE loss	Tversky loss	Weighted binary Cross-entropy
Full features	0.946393	0.851417	0.960879	0.738011	0.860874
Text + Rating	0.738170	0.738170	0.759645	0.738170	0.738170
Text + Sentiment	0.840750	0.738170	0.858569	0.738170	0.809689
Text only	0.738170	0.738170	0.758170	0.738170	0.738170

Fig. 9 ROC-AUC for training data

Once the model is trained with the initial dataset, the next critical step is to evaluate its performance on new, unseen data. To ensure a comprehensive assessment of the model's generalizability and robustness, it is tested using sample datasets from a diverse range of product categories including Cell Phones and Accessories, Digital Music, Electronics, Industrial and Scientific, and Software (see Table 5). By utilizing such a varied selection of categories, we can gauge the model's effectiveness across different types of products and determine its ability to accurately classify and analyze reviews from multiple sectors of the e-commerce landscape. Therefore, Table 6 summarizes the testing performances of the fake reviews. The metrics provided are the ROC-AUC score, Precision, Recall, and F1-Score, which collectively assess the model's accuracy and reliability in classifying fake reviews. For a better view of the ROC-AUC, see the appendix.

In the Cell Phones and Accessories category, the model achieved strong results, as evidenced by a ROC-AUC score of 0.93. With a Precision of 0.986, the model demonstrated high accuracy in identifying fake reviews, and a Recall of 0.97 indicated that it effectively detected most fake reviews in this category. The F1-Score, which stood at 0.978, highlighted the model's ability to maintain a well-balanced performance between Precision and Recall.

For the Digital Music category, the model achieved a ROC-AUC score of 0.92, indicating a high ability to differentiate between fake and non-fake reviews. The Precision of 0.981 suggests that the vast majority of reviews identified as fake are indeed fake, while a Recall of 0.963 indicates that the model successfully detected 96.3% of all actual fake reviews. The F1-Score of 0.972 demonstrates a balanced and robust performance.

Table 5 Summary of the testing datasets

Dataset	Fake	Non-fake
Cell phones and accessories	8418	68,582
Digital music	7630	50,545
Electronics	24,919	87,041
Industrial and scientific	1916	43,702
Software	5808	2739

Table 6 Accuracy of testing data: HACE loss function

Dataset	ROC-AUC	Precision	Recall	F1-score
Cell phones and accessories	0.93	0.986	0.97	0.978
Digital music	0.92	0.981	0.963	0.972
Electronics	0.93	0.968	0.955	0.961
Industrial and scientific	0.94	0.995	0.979	0.987
Software	0.86	0.954	0.74	0.834

In the Electronics category, the model performed similarly well, with a ROC-AUC score of 0.93. The Precision was 0.968, indicating strong accuracy in positive predictions, and the Recall was 0.955, showing effective detection of fake reviews. The F1-Score was 0.961, reflecting a good balance between Precision and Recall.

The highest performance was observed in the Industrial and Scientific category, with a ROC-AUC score of 0.94, Precision of 0.995, Recall of 0.979, and F1-Score of 0.987. This indicates an almost perfect classification capability in this category.

Conversely, the Software category showed the lowest performance among the groups, with a ROC-AUC score of 0.86. The Precision was still high at 0.954, but the Recall was lower at 0.74, indicating some difficulty in detecting all fake reviews. The F1-Score of 0.834, though lower than in other categories, still demonstrates reasonable effectiveness.

Overall, the table illustrates the CNN model's strong performance across different product categories, with particularly high accuracy and reliability in identifying fake reviews in the Industrial and Scientific category, while showing slightly reduced effectiveness in the Software category.

5 Conclusion and discussion

The exploratory data analysis provides critical insights into the characteristics and distribution of fake and non-fake reviews. As illustrated in Fig. 4, there exists an important overlap in the vocabulary used within both genuine and deceptive reviews. This lexical resemblance emphasizes the crafted nature of fake reviews, which are intentionally composed of mimicking authentic feedback. Such linguistic similarities imply that reliance solely on traditional keyword filtering techniques may fall short, thereby emphasizing the necessity for more sophisticated modeling approaches. By analyzing the cosine similarity between non-fake reviews and machine-generated reviews, a threshold of 0.76 is established, effectively classifying reviews based on their similarity scores. This threshold ensures that the classification process is based on a statistically significant measure of similarity, allowing for more accurate detection of fake reviews.

The detailed examination of ratings, user behavior, and review content revealed distinct patterns. According to Fig. 5, fake reviews frequently include high ratings, particularly five-star ratings, and are often longer in length compared to their non-fake counterparts. This lengthier nature, along with longer summaries, suggests that fake reviews tend to provide more elaborate or embellished content and are focused on 5 star ratings. Additionally, the sentiment analysis indicates that fake reviews generally exhibit more positive sentiment compared to non-fake reviews. These features—review length, sentiment score, and rating—emerge as significant indicators for distinguishing fake reviews from genuine reviews.

The application of a CNN for automated classification has proven highly effective, largely due to the strategic use of HACE as the loss function. This choice is particularly advantageous for handling imbalanced data, where the disparity between the number of fake and non-fake reviews can skew the model's performance. HACE addresses this imbalance by assigning dynamic weights to the classes case by case, thereby emphasizing the minority class (fake reviews) and reducing bias toward the majority class (non-fake reviews). Further, adding a smoothing parameter helps the HACE to learn from difficult cases. These adjustments ensure that the model gives more consideration to the often underrepresented fake reviews, leading to a more balanced and accurate classification. Adjusting offers on the fly gives us a real edge because it aligns the method with how the model

evolves over time. Instead of sticking with rigid parameters, we update the class weight penalties based on actual performance data, which helps us better handle imbalanced datasets and tricky classification challenges. Incorporating techniques like label smoothing and tweaking the focusing parameter makes the approach even more effective, especially when dealing with uncertain or borderline cases such as spotting fake reviews.

Training the model with critical features such as rating, review length, and sentiment scores, the CNN model achieved remarkable performance metrics. The use of convolutional and pooling layers makes it well-suited for extracting patterns from text and mix data types. The model attained a ROC-AUC score of 0.961, reflecting its near-perfect accuracy in distinguishing between fake and genuine reviews. The high precision of 0.96 indicates that the majority of reviews classified as fake are indeed fake, demonstrating the model's strong ability to minimize false positives. Concurrently, the high recall of 0.94 highlights the model's effectiveness in identifying a substantial portion of all actual fake reviews, thereby minimizing false negatives.

Proposed deep neural networks that combine word embeddings with emotion analysis by Hajek et al., reported about 82% accuracy for Amazon data and around 90% for hotel and restaurant reviews. In contrast, our CNN model achieved a much higher ROC-AUC of 96.1%, showing a notable improvement. The main difference is our innovative HACE loss function and the bootstrap-based method we used to set thresholds, which helped boost performance.

Further, our method achieves superior performance through our comprehensive feature engineering (review length, sentiment scores, ratings), the novel HACE loss function, and localized pattern recognition, while compare to Sun et al.'s proposed hybrid model, which combines CNN with two SVM models, and Zhang et al.'s Longformer models, which incorporates behavior-sensitive feature extractors via hybrid CNN-BiLSTM architectures.

Moreover, our findings suggest that detecting fake reviews in our dataset relies more on local language patterns, such as repeated adjectives or sudden shifts in sentiment, rather than complex sentence structures. This setup enables the CNN's layered approach to feature extraction to surpass the self-attention mechanisms typically used in transformer-based models (see Table 2). The small differences in precision (CNN: 0.931 versus transformers: 0.920 and 0.910) indicate that both models are good at minimizing false positives. Balancing the theoretical capabilities of the models with their effectiveness for this specific task justifies choosing the CNN, especially considering real-world review moderation scenarios where speed and hardware costs are essential factors.

VADER's main strength in this research is its ability to handle short, casual texts like online reviews reliably and accurately. It is lightweight, requiring minimal computational power and no need for large training datasets, which makes it a practical choice. However, a downside is that it sometimes struggles to correctly interpret sarcasm or irony in context, especially in complex or subtle instances, which can lead to occasional mislabeling. We chose to use pre-trained GloVe embeddings because they help reduce the amount of computing power needed and speed up the training process, unlike creating new embeddings from scratch. These embeddings worked quite well at first because they were trained on a sizable dataset, which helped them understand key word relationships that are valuable for analyzing Amazon reviews. However, a downside of using pre-trained embeddings is that some words specific to product reviews, such as unique jargon or brand names, might not be represented as accurately, which could lead to slightly lower precision when dealing with specialized product terms. Additionally, one of the main advantages of choosing TweetTokenizer was its ability to correctly process the informal language typical of our dataset, which leads to more accurate token representations than general or simpler tokenization methods.

The model's performance for unseen data is consistent across various product categories, with particularly notable success in the Industrial and Scientific category, where it excels with the highest accuracy. This performance underscores the robustness of the model in handling diverse review datasets. While the model shows slightly lower performance in the Software category, with a still robust accuracy, this variation suggests opportunities for further refinement and optimization. Overall, the effective use of proposed HACE and the model's strong performance across categories confirm its suitability and reliability for automating the detection of fake reviews, offering valuable insights for practical applications and further development.

Furthermore, our results clearly support the decision to use the GPT-2 language model. We achieved above 95% accuracy rate in correctly distinguishing between fake reviews generated by GPT-2 and authentic reviews. This high

level of accuracy suggests that the language patterns used by GPT-2 closely mirror those found in genuine deceptive reviews, allowing for meaningful analysis and effective detection. Therefore, using GPT-2 doesn't affect the reliability or accuracy of our study. However, it is prudent to acknowledge that future research should consider evaluating whether the latest advancements in language models, such as GPT-3, GPT-4, LLaMA, or Claude, could further enhance the accuracy of detecting fake reviews and improve the quality of synthetically generated data. Exploring these advanced models holds major potential for increasing linguistic authenticity and enriching contextual understanding. Such developments open promising avenues for future studies and technological innovations within this domain.

We carefully created this CNN architecture to strike a balance between complexity, speed, and accuracy. Its layered convolutional design is able to pick up on language patterns, from simple word groups like n-grams to deeper meanings. We tinkered with kernel sizes and filters based on real-world testing to get the best results for classifying reviews. Using ReLU activation functions and pooling layers helps the model run efficiently and pulls out important features, while Dropout was added to keep it from overfitting, especially since our dataset is not enormous but still quite detailed. In summary, these design choices together achieved the highest practical performance, making the model reliable and effective for real-world fake review detection tasks.

Our innovative application leverages advanced machine learning models to detect fake reviews on Amazon, offering a unique approach to maintaining e-commerce integrity. By incorporating Amazon's own Voice of the Customer metrics, including Verified Purchase (VP) and Non-Verified Purchase (NVP) classifications, our approach provides a nuanced and context-aware analysis of review authenticity. This powerful solution is designed for versatility, capable of running as a standalone application on personal computers or seamlessly integrating into web browsers as an extension in future. What sets our software apart is its robust statistical foundation, ensuring reliable and accurate results without relying on external data sources. Users can harness the power of this tool using only the data already at their disposal, making it both privacy-conscious and immediately actionable. By combining cutting-edge algorithms with user-friendly implementation, our software stands as a formidable asset in the ongoing effort to preserve trust and transparency in online marketplaces. Whether you're a consumer seeking authentic product insights or a platform moderator aiming to maintain review integrity, our tool offers a sophisticated yet accessible solution to the challenge of fake review detection.

Overall, the combination of feature analysis and advanced model training underscores the effectiveness of the CNN approach in classifying fake reviews. The identified features such as review length, sentiment score, and rating are instrumental in building a reliable and accurate model, showcasing its potential for real-world application in automating the detection of fake reviews across diverse product categories.

Appendix

See Table 7 and Figs. 10, 11 and 12.

Table 7 Accuracy of the CNN model for the optimum values of parameters α and β in weighted binary cross-entropy

α	β	ROC-AUC
0.2	1.8	0.9441
0.4	1.6	0.9437
0.6	1.4	0.9436
0.8	1.2	0.9418
1	1	0.9394
1.2	0.8	0.9379
1.4	0.6	0.9394
1.6	0.4	0.9382
1.8	0.2	0.9408
2	0	0.9389

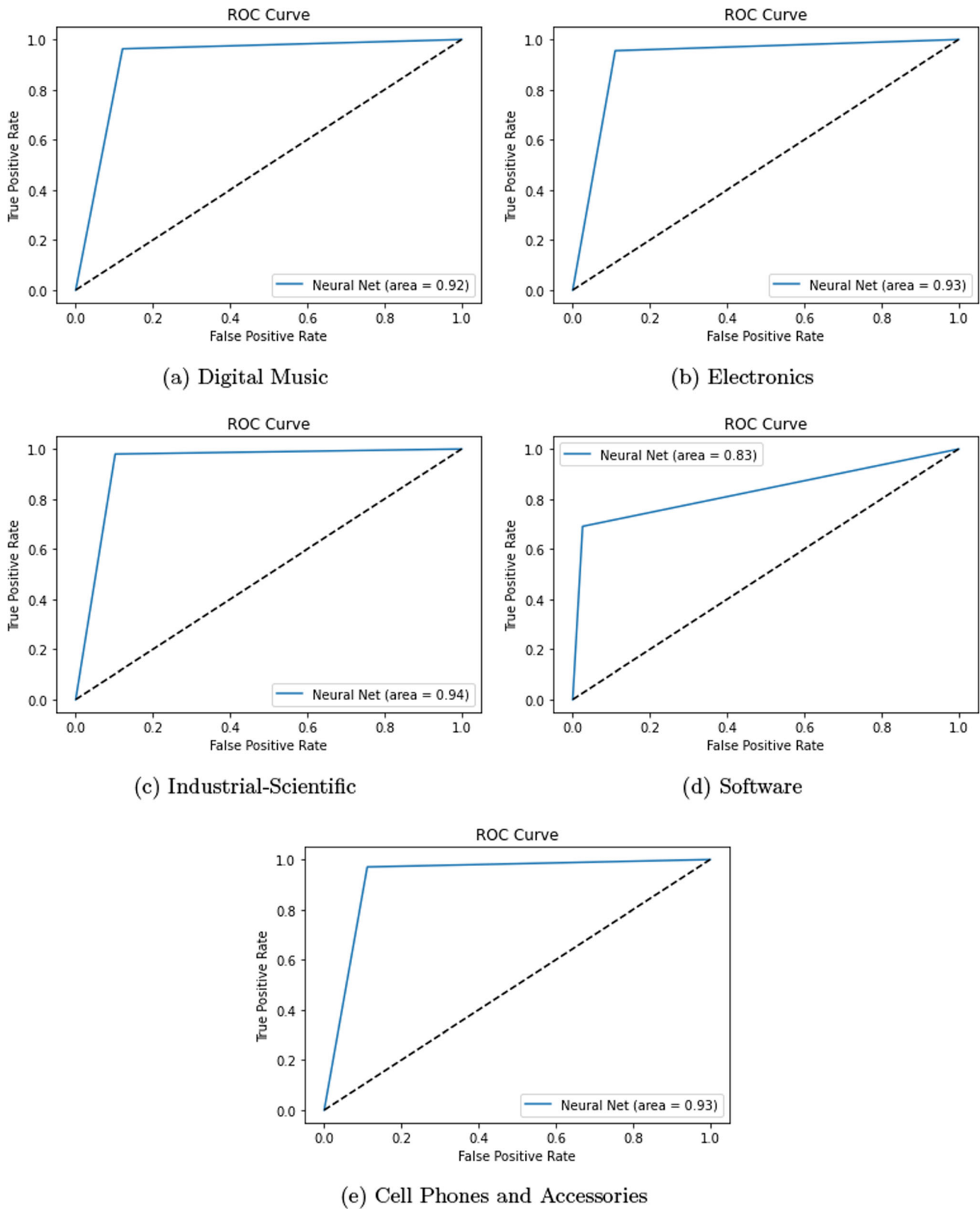


Fig. 10 Graphical representation of ROC-AUC scores for each test dataset

Fig. 11 E_{HACE} loss changes regarding the α and γ parameter values

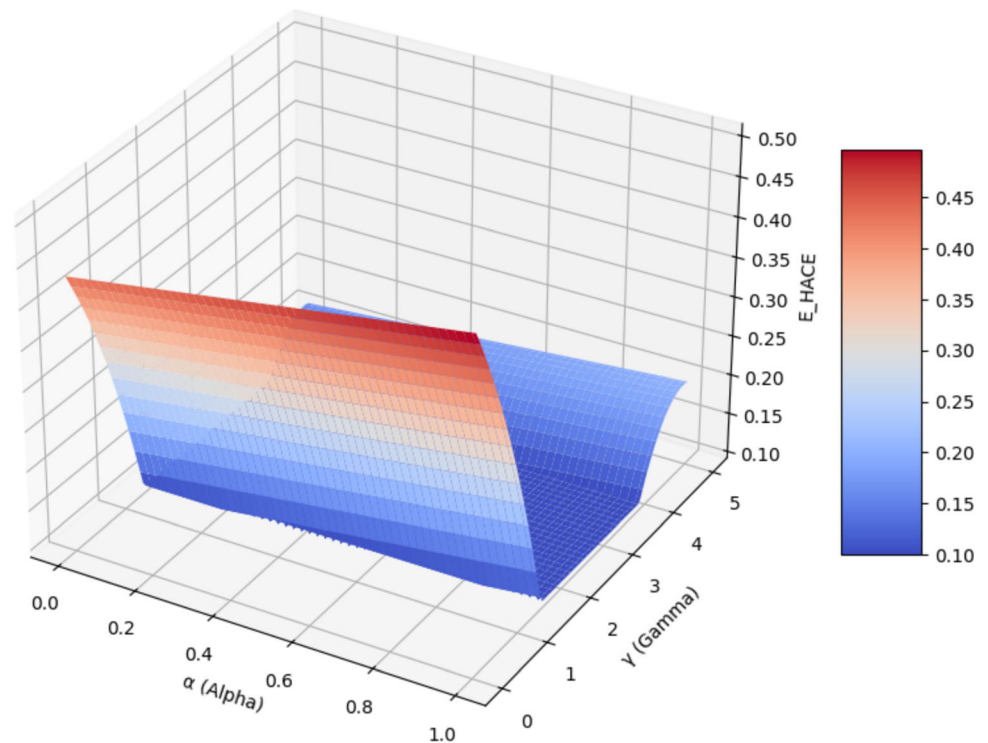
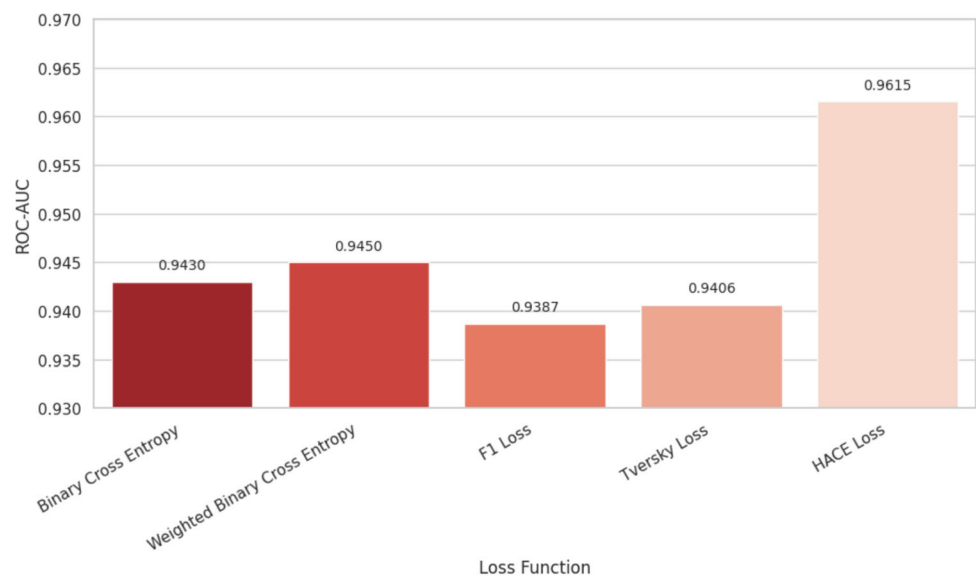


Fig. 12 ROC_AUC performance for different loss functions



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Data availability All datasets are publicly available and sources: Justifying recommendations using distantly labeled reviews and fined-grained aspects, Jianmo Ni, Jiacheng Li, Julian McAuley Empirical Methods in Natural Language Processing (EMNLP), 2019. Moreover, some datasets and further details for reproducibility can be found in the project's GitHub page: [AmazonReviews.git](https://github.com/AmazonReviews)

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

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